

Up until now, you have likely just used the Standard Cartesian Coordinate system where everything is based on squares and

(x, y) refers to a horizontal and vertical position.

Newton introduced a different system that is more convenient in a variety of applications.

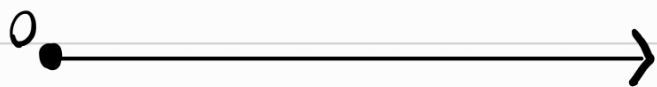
What Is the Polar Coordinate System?

Take a point and a line from this point, for convenience, make this line horizontal.

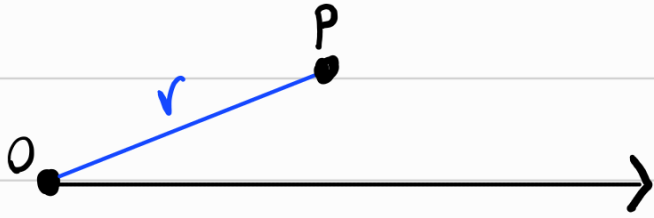
Call this point the "pole" or "origin" and the line the "polar axis"



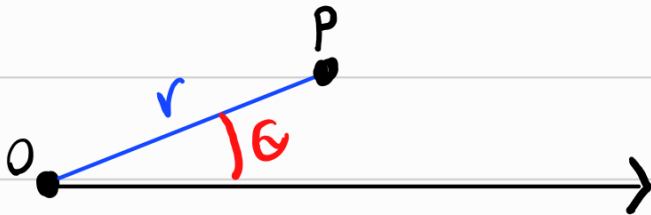
Now, anywhere, place another point P.



Relative to the origin, this point is some positive distance, r , away



However, there are a lot of points distance r from the origin. Unique to P , we take the angle θ , from the polar axis to the segment $\overline{OP}$



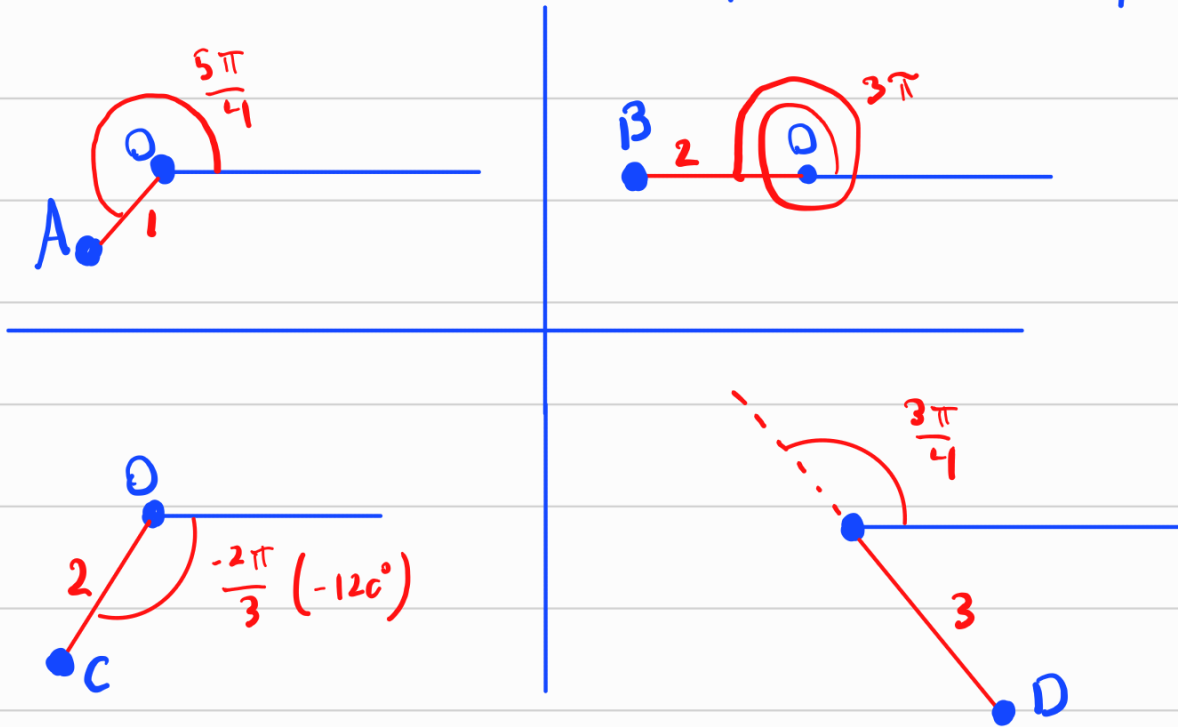
The values (r, θ) uniquely determine P .

So, every point in Polar Space is written as a pair

$$(r, \theta)$$

Ex: The Points

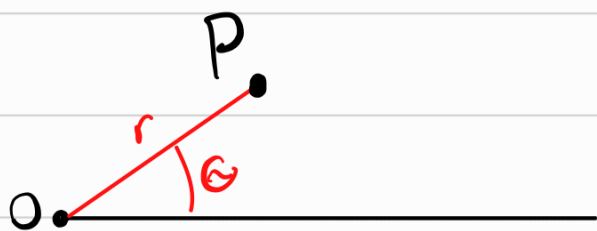
$$A(1, \frac{5\pi}{4}), B(2, 3\pi), C(2, -\frac{2\pi}{3}), D(-3, \frac{3\pi}{4})$$



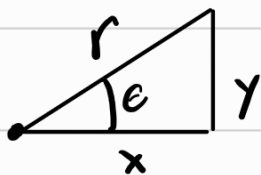
Relation between Polar and Cartesian

Let's find a way to go back and forth from Polar and Cartesian.

If $P(r, \theta)$ is a polar point, we have the following picture



drawing a triangle



we can see that

$$x = r \cos \theta, \quad y = r \sin \theta$$

and,

$$r^2 = x^2 + y^2, \quad \tan(\theta) = \frac{y}{x} \leadsto \theta^* = \arctan\left(\frac{y}{x}\right)$$

$$\ast \arctan(z) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

so we need to be careful.

Ex:

$(2, \pi/3)$ in Cartesian Coordinates

is

$$x = 2 \cos(\pi/3) = 2 \cdot \frac{1}{2} = 1$$

$$y = 2 \sin(\pi/3) = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$

$$\boxed{(1, \sqrt{3})}$$

Ex: The Cartesian point $(1, -1)$ is Polar

$$r^2 = x^2 + y^2$$

$$\tan \theta = \frac{-1}{1} = -1$$

$$r = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \frac{-\pi}{4} \text{ or } \frac{3\pi}{4}$$

Quadrant IV Quadrant II

Since, $(1, -1)$ is in QIV, $\theta = \frac{-\pi}{4}$ (or $\frac{7\pi}{4}$)

So, in Polar

$$(1, -1) \rightarrow \boxed{\left(\sqrt{2}, \frac{-\pi}{4}\right)} \text{ or } \left(\sqrt{2}, \frac{7\pi}{4}\right)$$

Graphs of Polar Curves

These are functions $g(r) = f(\theta)$, or more general $F(r, \theta) = 0$ (i.e. some combination of r, θ , and some operators).

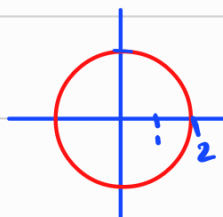
To plot the graph of a Polar function, we want to think of all pairs (r, θ) that satisfy the relation.

★ This is quite difficult to do ★

Ex:

$$r = 2$$

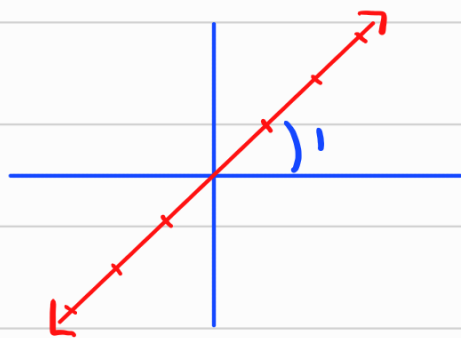
here θ doesn't matter. So, we need all points of a distance 2 from the origin. This describes a circle of radius 2



Ex:

$$\theta = 1$$

Always remember θ is measured in radians,
So this describes all points of any radius
at angle 1 rad. This is a line



Sometimes it is easier to convert things to
Cartesian equations to get a better understanding

Ex:

$$r = 2 \cos \theta$$

Recall $r^2 = x^2 + y^2$, $x = r \cos \theta$, $y = r \sin \theta$

So

$$r = 2 \cos \theta = 2 \frac{x}{r}$$

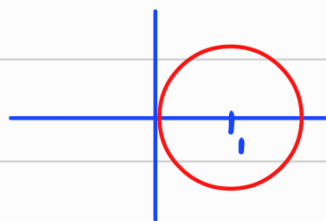
$$r^2 = 2x$$

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$\rightarrow x^2 - 2x + 1 - 1 + y^2 = 0$$

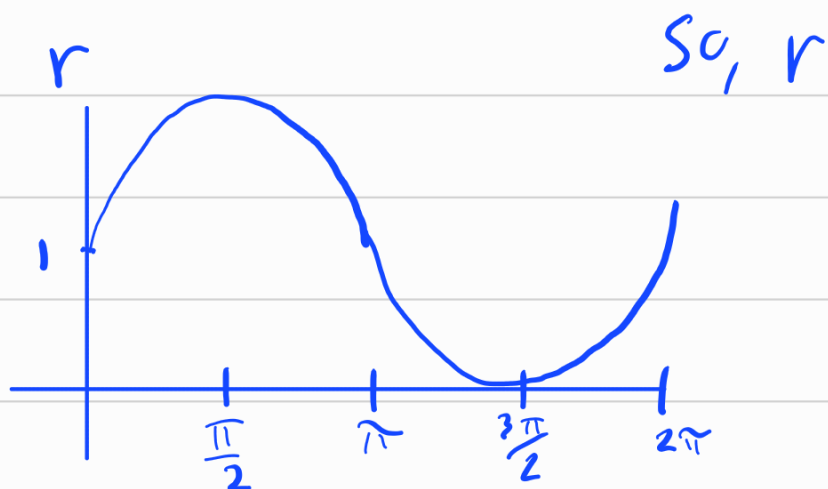
$$(x-1)^2 + y^2 = 1$$



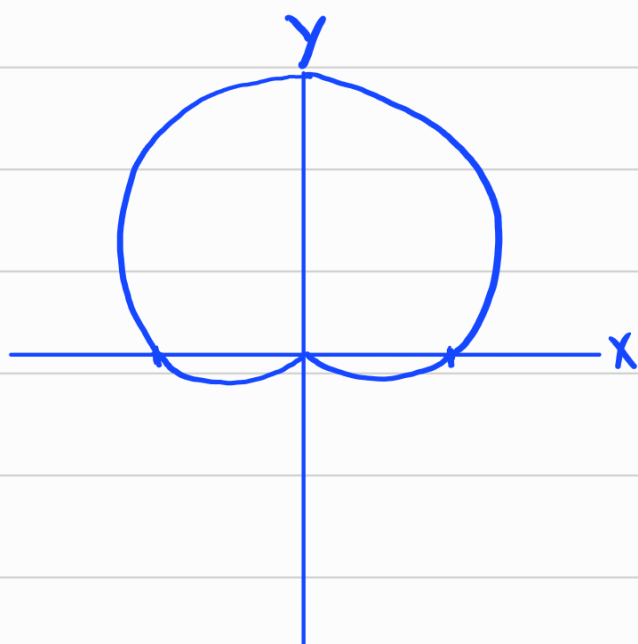
Another way to plot Polar functions is to solve for r and plot r vs. θ to gain an idea of the movement of a point.

Ex:

$$r = 1 + \sin \theta$$



- Starts at 1
- Increases to 2 at $\frac{\pi}{2}$
- decreases to 1 at π
- decreases to 0 at $\frac{3\pi}{2}$
- Increases to 1 at 2π

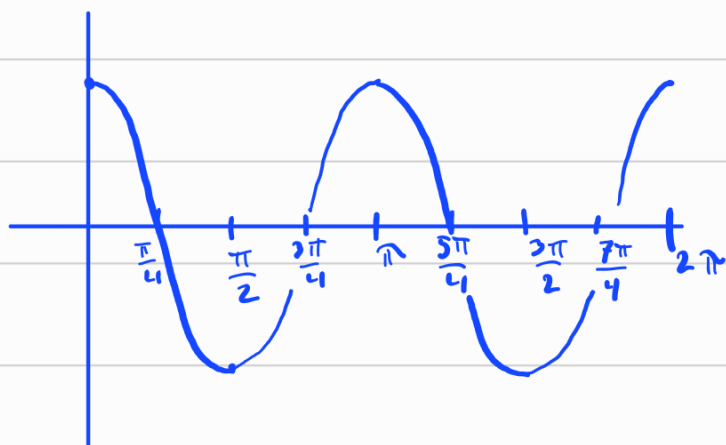


~ This shape is called a "cardioid" due to its heart like shape

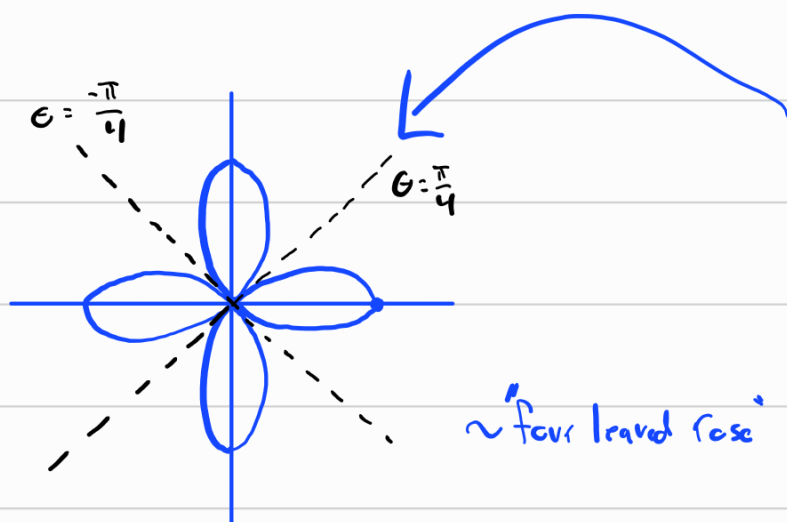
Ex:

$$r = \cos(2\theta)$$

So r ,



- is 1 at 0
- decreases to 0 at $\frac{\pi}{4}$
- decreases to -1 at $\frac{\pi}{2}$
- increases to 0 at $\frac{3\pi}{4}$
- increases to 1 at π
- repeats



There are some nice symmetry properties to help with graph.

a) If the polar equation is unchanged when you replace θ with $-\theta$. The graph is symmetric over the polar axis (horizontal axis)

Ex: $r = 2\cos\theta$

b) If the equation is unchanged when r is replaced by $-r$, or when θ is replaced by $\theta + \pi$, the curve has 180° rotational symmetry (symmetric around origin)

Ex: $r = \cos(2\theta)$

c) If the equation is unchanged by $\theta = \pi - \theta$, the curve has symmetry around the vertical line $\theta = \frac{\pi}{2}$.

Ex: $r = 1 + \sin\theta$ (recall $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$)